Motion Gating versus Signal Trust Scores for Wearable Heart-Rate Alerting: A Pre-Specified Fault-Injection Benchmark

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***Abstract*—Continuous wrist photoplethysmography is prone to motion artifacts, optical decoupling, and telemetry drops that trigger spurious resting tachycardia alarms. While digital health platforms routinely deploy multi-signal software quality metrics to suppress false alarms, the incremental utility of software trust scoring over simple kinematic motion gating is rarely benchmarked under controlled protocols. We present a reproducible benchmark evaluating six causal alerting arms across fifteen ambulatory subjects from the public PPG-DaLiA dataset under a pre-specified, version-tagged protocol without an external public registry. To address natural tachycardia sparsity in healthy individuals, labelled synthetic resting tachycardia episodes were injected into derived pulse-rate streams alongside two seeded telemetry fault regimes and high-motion optical noise. Pre-specified hypothesis estimation found that kinematic stillness gating accounted for the largest share of false-alert suppression, whereas multi-attribute software trust scoring yielded no statistically significant incremental benefit across any evaluated condition. The no-motion trust ablation alone reduced false alerts by about half relative to personal baselines. Furthermore, conservative software abstention degraded event sensitivity under optical noise (exploratory, driven by two subjects). Inductive conformal abstention, pre-specified in Amendment 3, implemented after the other arms' results were seen, achieved the lowest false-alert frequency at the expense of recall. All findings are derived from synthetic episodes in healthy volunteers and carry no clinical standing.**

***Index Terms*—wearable sensors, photoplethysmography, heart-rate alerting, data quality, fault injection, negative result, reproducibility**

I. Introduction

Ambulatory continuous monitoring with consumer and medical-grade wearables offers substantial potential for early arrhythmia detection. However, wrist photoplethysmography (PPG) is acutely sensitive to motion artifacts, optical decoupling, transmission dropouts, and flatline sensor freezes. When raw optical or derived telemetry signals degrade, alerting algorithms frequently emit spurious notifications. In wearable monitoring and clinical telemetry alike, frequent false alerts risk causing user annoyance and alert fatigue, which can compromise long-term adherence.

To mitigate spurious alarms, modern digital health architectures deploy signal quality indices (SQIs) and multi-component trust heuristics that penalize missing samples, flag frozen values, and bound rate-of-change dynamics. Despite their widespread adoption, the isolated incremental utility of multi-component software trust scores over straightforward kinematic stillness gating (e.g., Euclidean Norm Minus One, ENMO) has rarely been benchmarked under controlled cross-validation. The core protocol was version-tagged before the alerting experiments; later amendments are listed in Table VI; no external public registry was used.

This paper reports a pre-specified fault-injection benchmark for wearable heart-rate alerting. Our contributions are:

1) **Open Benchmarking Framework:** A reproducible pipeline formalizing seven fault types plus a clean condition (clean, gaps, stale values, duplicates, spikes, drift, scaling errors, and raw optical noise) evaluated across 35.94 total recording hours (31.27 non-event exposure hours) of multi-activity ambulatory recordings.
2) **Causal Leave-One-Subject-Out (LOSO) Evaluation:** Six distinct alerting arms evaluated under strict data quarantine, separating training folds from held-out evaluation subjects.
3) **Rigorous Negative Finding:** Empirical evaluation of pre-specified estimation hypothesis H_1, demonstrating that kinematic motion gating delivers 67.9%–74.2% false alert reduction relative to the ungated personal baseline, whereas multi-attribute software trust scoring provides no statistically detectable incremental benefit ($p \geq 0.28$).

II. Literature Survey

A. Motion Artifact Mitigation in Wrist PPG: Wrist photoplethysmography (PPG) optically monitors pulsatile blood volume changes in the cutaneous microvascular bed via illuminated LEDs and paired photodiodes [3], [7]. In resting conditions, PPG tracks pulsatile cardiac cycles with high morphological fidelity. However, in ambulatory environments, physical movement introduces severe motion artifacts (MA) caused by displacement between the sensor photodiode and

the skin, venous blood sloshing, and microvascular pressure transients [7], [8]. These movement-induced spectral peaks overlap directly with cardiac fundamental frequencies (1.0–3.5 Hz, corresponding to 60–210 bpm), severely biasing unconstrained peak-detection and Fourier pulse-rate estimates [9], [10]. To recover corrupted cardiac signals, multi-channel kinematic fusion has emerged as a cornerstone of wearable signal processing. Zhang *et al.* [9] introduced the TROIKA framework, combining temporal difference filtering, sparse signal reconstruction, and peak tracking against triaxial accelerometry. Salehizadeh *et al.* [10] demonstrated time-varying spectral filtering to isolate motion harmonics during treadmill exercises. Reiss *et al.* [1] introduced the 15-subject PPG-DaLiA benchmark and deep convolutional tracking architectures, establishing standard Mean Absolute Error (MAE) baselines against chest ECG while noting performance bottlenecks during non-rhythmic wrist movements. Furthermore, Bent *et al.* [8] systematically characterized device-level sources of wearable optical error across commercial monitors, confirming physical motion as the primary driver of tracking inaccuracies.

B. Signal Quality Indices and Multi-Modal Fusion: Because signal reconstruction cannot fully recover severely decoupled waveforms, extensive research has focused on automated Signal Quality Indices (SQIs) to identify and discard unreliable epochs [2], [3]. Elgendi [2] surveyed and evaluated eight distinct morphological and statistical SQIs, highlighting skewness, kurtosis, and perfusion indices on stationary recordings. Karlen *et al.* [11] advanced real-time quality estimation using repeated Gaussian filtering and continuous template cross-correlation, demonstrating robust identification of uninterpretable pulse waves in pediatric and ambulatory settings. Selvaraj *et al.* [12] proposed statistical stationarity testing for automated motion-noise detection in wearable PPG. In wireless patient monitoring, cross-channel telemetry fusion enhances alert fidelity. Orphanidou *et al.* [13] developed unified ECG and PPG quality metrics combining envelope analysis, template matching, and physiological bounds, validating utility on ambulatory wireless recordings. Li and Clifford [14] demonstrated that fusing multi-parameter signal quality metrics with dynamic Bayesian estimation drastically suppresses false alarms in intensive care telemetry without compromising clinical event sensitivity.

C. Alarm Fatigue, Telemetry, and Conformal Uncertainty: Frequent false alerts from wearable and bedside monitors impose severe clinical and behavioral costs. In hospital settings, Drew *et al.* [15] and Cvach [16] documented pervasive monitor alarm fatigue, where false alarm rates exceeding 85% desensitize clinicians and degrade response times to life-threatening arrhythmias. Clifford *et al.* [17] formalized this challenge in the PhysioNet/CinC 2015 Benchmark, showing that algorithmic false-alarm suppression requires joint multi-modal cross-checking. At consumer scale, the Apple Heart Study ($N = 419{,}297$) [18] confirmed that optical pulse irregularity can trigger widespread clinical escalations, demonstrating the critical necessity of stringent optical quality gates prior to alerting. Under non-stationary ambulatory noise, deterministic predictions lack distribution-free confidence. Conformal prediction [19], [20] provides finite-sample, distribution-free coverage guarantees with valid non-conformity calibration. Vazquez and Facelli [4] reviewed conformal prediction across fourteen clinical applications, demonstrating that selective abstention preserves decision purity in high-stakes healthcare scenarios. Table I compares this benchmark against opened and verified literature sources.

TABLE I
RELATED WORK COMPARISON ON OPENED AND VERIFIED LITERATURE

Study	Cohort	Motion Gate	Trust/SQI	Scope & Benchmark Focus
Zhang *et al.* (2015) [9]	$N = 12$ treadmill	ACC / SBL	Spectral	Sparse reconstruction for intensive physical exercise HR.
Reiss *et al.* (2019) [1]	$N = 15$ PPG-DaLiA	ACC features	None	Deep PPG CNN tracking; MAE metric; no alerting benchmark.
Elgendi (2016) [2]	Static finger PPG	None	8 SQIs	Peak detection under stationary clean/noisy segments.
Orphanidou *et al.* (2015) [13]	Clinical & ambul.	None	Multi-SQI	Morphological ECG/PPG quality indexing for wireless telemetry.
Clifford *et al.* (2016) [17]	PhysioNet CinC'15	Multi-modal	Multi-SQI	Multi-lead ICU alarm reduction benchmark ($N = 1250$).
Charlton *et al.* (2023) [3]	Review	Varied	Review	Wearable roadmap; highlights benchmark datasets, SQIs, standards.
Vazquez & Facelli (2022) [4]	14 clinical studies	N/A	Conformal	Conformal guarantees in clinical prediction tasks.
This Benchmark	$N = 15$ ambulatory	ENMO_{10s}	Multi-SQI	Causal alerting Arms 1–4; 7 fault types + clean; LOSO.

III. DATASET AND PROTOCOL

A. PPG-DaLiA Dataset

We evaluate on the public PPG-DaLiA benchmark (UCI ID 495, CC BY 4.0) [1], comprising 15 healthy subjects undergoing 2.5 hours of authentic daily activities (Table II). Reference heart rates are derived from gold-standard chest ECG. Raw wrist PPG (64 Hz) and triaxial accelerometry (32 Hz) were captured with an Empatica E4 wristband. Strict quarantine was maintained: chest ECG reference was isolated solely for ground truth evaluation and was inaccessible to alerting arms.

B. Labelled Synthetic Tachycardia Injection

Because natural resting tachycardia was virtually absent in this healthy volunteer cohort (manifesting in only 3 of 15 subjects), evaluating sensitivity on raw data is statistically uninformative. Per Amendment 2, we injected 112 labelled synthetic resting tachycardia episodes into derived pulse-rate streams during verified low-motion windows ($\mathrm{ENMO}_{10s} \leq 0.05\,\mathrm{g}$). Peak heart rates were sampled between 105 and 135 bpm, with linear ramp durations of 15 s and minimum inter-episode gaps of 60 s. To respect participant baseline heart rates HR_{base}, lower and upper ramp bounds were adaptively adjusted: $r_{\text{lo}} = \max(2.0, (105 - HR_{\text{base}})/15)$, $r_{\text{hi}} = \min(4.0, (135 - HR_{\text{base}})/15)$, with plateau peak heart rate clamped to $[105.0, 135.0]$ bpm.

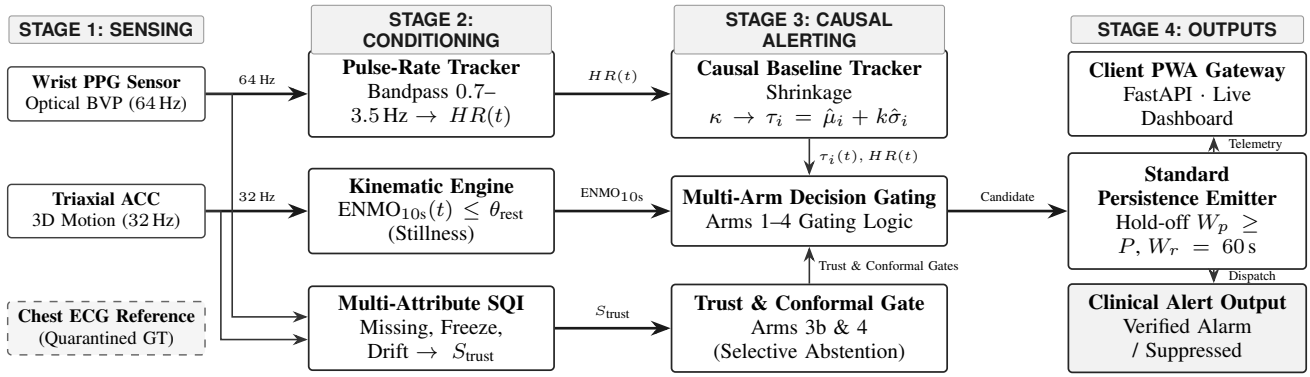


Fig. 1. System Architecture: End-to-end wearable telemetry, signal conditioning, causal baseline estimation, multi-arm gating, and standard persistence emitter schematic.

C. Seeded Fault Regimes

Telemetry faults were simulated under two pre-specified schedules alongside an optical corruption condition (Table II): Condition 0 (clean background, 31.27 h non-event exposure); Regime A (low duty, $\approx 10\%$ faulted time); Regime B (nominal duty, $\approx$ 25–30% faulted time) injecting six deterministic fault types (F1 gaps, F2 frozen flatlines, F3 duplicates, F4 spikes, F5 drift, F6 scale errors; seed 42); and F7 high-motion optical noise (additive Gaussian noise continuously modulated by wrist 3D acceleration injected into raw 64 Hz BVP at SNR $+5.0$ dB prior to spectral estimation).

TABLE II
PPG-DaLiA Benchmark Dataset & Injected Fault Regimes

Part A: PPG-DaLiA Benchmark Dataset Characteristics [1]	
Subjects	$N = 15$ healthy individuals (7 male, 8 female, age 21–55)
Duration	≈ 2.4 h per subject (35.94 total recording hours; 31.27 non-event exposure hours)
Activities	Sitting, Walking, Stairs, Cycling, Driving, Lunch, Soccer
Sensors	Wrist Empatica E4 (BVP: 64 Hz, 3D ACC: 32 Hz)
Ground Truth	Chest ECG (RespiBAN, 700 Hz, strictly quarantined)
Part B: Evaluated Telemetry & Sensor Fault Regimes (Amendment 3)	
Condition 0	Clean ambulatory background (31.27 h non-event exposure)
Regime A	Low-duty ($\approx$ 10% faults): sporadic gaps (F1), freeze (F2), duplicates (F3), drift (F5), scale errors (F6)
Regime B	Nominal-duty ($\approx$ 25–30% faults): prolonged flatlines, frequent dropouts, scale errors (seed 42)
F7 Noise	Additive Gaussian noise continuously modulated by wrist 3D acceleration injected into raw BVP (SNR $+5$ dB)

IV. System Architecture

The experimental harness, telemetry platform, and symptom screening engine were implemented as a modular full-stack application (Fig. 1). The React 18 PWA frontend features live telemetry streaming and an interactive LOSO dashboard. Functional specifications are operational: Requirement FR-04 enforces automated data-quality exclusion; Module FR-06 implements guided symptom assessment with mandatory answer review; and Module FR-07 provides local feature attributions ("DEMO RULESET v2.1, not evaluated"). The FastAPI/SQLite backend incorporates the `vitasense.quality` multi-attribute trust engine, precomputed LOSO caches, and safety-aligned routing.

V. Methods and Algorithms

A. Evaluated Alerting Arms

We benchmark six causal alerting arms without future peeking (Table III):

TABLE III
Evaluated Alerting Arms, Decision Rules & Tuning Grids

Alerting Arm	**Decision Rule & Tuning Grid**	**Hold-off R**
Arm 1: Fixed Pop.	$HR(t) > \tau_{\text{pop}}$ $\tau_{\text{pop}} \in \{100, 110, 120, 125, 130\}$ bpm, $P \in \{10, 20, 30\}$ s	60 s
Arm 2: Personal	$HR(t) > \max(\tau_{\text{flr}}, \hat{\mu}_i(t) + k \cdot \hat{\sigma}_i(t))$ $k \in \{2.0, \ldots, 4.0\}$, $\kappa \in \{0.5, 1, 2\}$ h, $P \in \{10, 20, 30\}$ s	60 s
Arm 3a: +Kinematic	Arm 2 $\wedge$ $(\text{ENMO}_{10s}(t) \leq \theta_{\text{rest}})$ $\theta_{\text{rest}} \in \{0.03, 0.05, 0.08\}$ g, $P \in \{10, 20, 30\}$ s	60 s
Arm 3b: +Trust	Arm 2 $\wedge$ $(S_{\text{trust}}(t) \geq \theta_{\text{trust}})$ $\theta_{\text{trust}} \in \{0.60, 0.70, 0.80\}$, $P \in \{10, 20, 30\}$ s	60 s
Arm 3b-noMot (Motion Ablated)	Arm 2 $\wedge$ $(S_{\text{gap}} S_{\text{stale}} S_{\text{phys}} \geq \theta_{\text{trust}})$ $\theta_{\text{trust}} \in \{0.60, 0.70, 0.80\}$, $P \in \{10, 20, 30\}$ s	60 s
Arm 4: Conformal (pre-specified in Amend. 3, impl. after other arms' results seen)	Arm 2 $\wedge$ $\neg\text{Abstain}(t)$ $\alpha \in \{0.05, 0.10\}$, $\lambda \in \{0.5, 1.0\}$, $\theta_{\text{unc}} \in \{0.2, 0.3\}$	60 s

(1) **Arm 1 (Fixed Pop.):** Standard clinical threshold $\tau_{\text{pop}} \in \{100, 110, 120, 125, 130\}$ bpm; (2) **Arm 2 (Personal Baseline):** Dynamic cutoff $\tau_i(t) = \max(\tau_{\text{floor}}, \hat{\mu}_i(t) + k \cdot \hat{\sigma}_i(t))$ regularized toward training priors $(\mu_{\text{pop}}, \sigma_{\text{pop}})$ via empirical Bayes shrinkage κ (Algorithm 1); (3) **Arm 3a (Kinematic Gating):** Arm 2 baseline with stillness gate: alerts trigger if $HR(t) > \tau_i(t)$ and $\text{ENMO}_{10s}(t) \leq \theta_{\text{rest}}$; (4) **Arm 3b (Trust Score Gating):** Arm 2 gated by software trust: alerts trigger if $HR(t) > \tau_i(t)$ and $S_{\text{trust}}(t) \geq \theta_{\text{trust}}$; (5) **Arm 3b-noMotion (Trust Ablation):** Arm 3b with motion ablated ($w_{\text{mot}} = 0$); and (6) **Arm 4 (Conformal Abstention, pre-specified in Amendment 3, implemented after the other arms' results were seen):** Arm 2 baseline with inductive

conformal calibration on resting windows ($\alpha \in \{0.05, 0.10\}$, selecting $\alpha = 0.10$ across all 15 folds); abstains when nonconformity or multi-attribute trust defects exceed calibrated thresholds.

Algorithm 1 Causal Heart-Rate Alerting Pipeline with Motion Gating and Conformal Abstention

Require: Streaming pulse rate $HR[t]$, motion ENMO$[t]$, priors $(\mu_{\text{pop}}, \sigma_{\text{pop}})$, parameters $(k, \kappa, \theta_{\text{rest}}, \theta_{\text{trust}}, \alpha, \lambda, \theta_{\text{unc}}, P, R)$
Initialize: $c \leftarrow 0$, running mean $\bar{x} \leftarrow \mu_{\text{pop}}$, running std $s \leftarrow \sigma_{\text{pop}}$, $run_len \leftarrow 0$, $ref_cnt \leftarrow 0$
for each window $t = 0, 1, \ldots, N-1$ **do**
  Update Resting Baseline:
  if ENMO$[t] \le 0.05$ g $\wedge$ finite$(HR[t])$ **then**
    $c \leftarrow c + 1$; Update sample mean $\bar{x}$ and sample standard deviation s
    Shrinkage weight: $w \leftarrow \frac{c \cdot \Delta t / 3600}{c \cdot \Delta t / 3600 + \kappa}$
    $\hat{\mu}_i[t] \leftarrow w\bar{x} + (1-w)\mu_{\text{pop}}$; $\hat{\sigma}_i[t] \leftarrow ws + (1-w)\sigma_{\text{pop}}$
  end if
  Causal threshold: $\tau_i[t] \leftarrow \max(100.0, \hat{\mu}_i[t] + k \cdot \hat{\sigma}_i[t])$
  Compute Quality and Gating Masks:
  $M_{\text{kin}}[t] \leftarrow \mathbb{I}(\text{ENMO}_{10\text{s}}[t] \le \theta_{\text{rest}})$
  $S_{\text{trust}}[t] \leftarrow \text{clip}(S_{\text{gap}} \cdot S_{\text{stale}} \cdot S_{\text{phys}} \cdot S_{\text{mot}}, 0.0, 1.0)$; $M_{\text{trust}}[t] \leftarrow \mathbb{I}(S_{\text{trust}}[t] \ge \theta_{\text{trust}})$
  Augmented uncertainty: $U_{\text{tot}}[t] \leftarrow (\hat{\sigma}_i[t]/\hat{\mu}_i[t]) + \lambda(1 - S_{\text{trust}}[t])$
  Abstain condition: $Abstain[t] \leftarrow (\tau_i[t] \in [\hat{\mu}_i[t] \pm \hat{q}_{1-\alpha}\hat{\sigma}_i[t]]) \vee (U_{\text{tot}}[t] > \theta_{\text{unc}})$
  Evaluate Active Alerting Arm Trigger Condition $C[t]$**:**
  case Arm 1: $C[t] \leftarrow (HR[t] > \tau_{\text{pop}})$
  case Arm 2: $C[t] \leftarrow (HR[t] > \tau_i[t])$
  case Arm 3a: $C[t] \leftarrow (HR[t] > \tau_i[t]) \wedge M_{\text{kin}}[t]$
  case Arm 3b: $C[t] \leftarrow (HR[t] > \tau_i[t]) \wedge M_{\text{trust}}[t]$
  case Arm 4: $C[t] \leftarrow (HR[t] > \tau_i[t]) \wedge \neg Abstain[t]$
  Standard Emitter with Persistence and Refractory Hold-off:
  if $ref_cnt > 0$ **then**
    $ref_cnt \leftarrow ref_cnt - 1$; $Alert[t] \leftarrow$ FALSE
  else if $C[t]$ is TRUE **then**
    $run_len \leftarrow run_len + 1$
    if $run_len \ge P/\Delta t$ **then**
      $Alert[t] \leftarrow$ TRUE; $ref_cnt \leftarrow R/\Delta t$; $run_len \leftarrow 0$
    end if
  else
    $run_len \leftarrow 0$; $Alert[t] \leftarrow$ FALSE
  end if
end for
return Alert stream $Alert$

B. Shared Standard Alert Emitter

To ensure rigorous cross-arm comparability, all arms feed instantaneous binary triggers into a shared `StandardAlertEmitter` (Algorithm 1). Alerts fire if and only if triggering conditions hold continuously for $W_p = P/\text{step}$ windows ($P \in \{10, 20, 30\}$ s). Emitted alerts enforce a mandatory refractory hold-off of $W_r = R/\text{step}$ windows ($R = 60$ s).

VI. Experimental Setup

A. Leave-One-Subject-Out Cross-Validation & Scoring

Hyperparameters were tuned strictly within 15 LOSO training folds. For test subject $s \in \{1 \ldots 15\}$, candidate vectors were evaluated across the 14 training subjects pooled. The pre-specified rule selects parameters achieving $\ge 85\%$ training recall while minimizing false alerts per hour (FA/hr); if unreachable, the pre-specified fallback rule selects the setting maximizing training recall.

Scoring assesses event-level recall (injected episodes within $[t_{\text{start}}, t_{\text{end}}]$) alongside alert-level clean false alerts per hour (FA/hr), precision, and F_1 score. Paired differences (Δ = Arm 3b − Arm 3a) are tested via two-sided Wilcoxon signed-rank tests ($\alpha = 0.05$), Hodges-Lehmann median estimators, and 95% subject-bootstrap confidence intervals (10,000 resamples).

VII. Results and Discussion

A. Primary Hypothesis H_1 Outcome: Estimation Finding

In Protocol Amendment 3, the primary research question was formulated as an estimation question: "*What is the difference in False Alerts per Hour between Arm 3b and Arm 3a* (Δ*FA/hr* $=$ *FA/hr*$_{\text{Arm 3b}}$ − *FA/hr*$_{\text{Arm 3a}}$)*, quantified with a 95% subject-bootstrap confidence interval?*" with stated targets ($\ge 60\%$ reduction in FA/hr relative to kinematic-only gating, and $\ge 85\%$ recall on genuine synthetic alert events maintained as stated targets, not rigid pass/fail criteria). In paired comparisons (Table IV, Table V, Fig. 2b), Arm 3b demonstrated no statistically detectable false-alert reduction over Arm 3a across any condition:

specifically: Condition 0 (Clean) mean difference $+0.13$ FA/h (95% CI $[-0.54, +0.89]$, $W = 31.0$, $p = 0.86$); Regime A ($\approx 10\%$ faults) $+0.05$ FA/h (95% CI $[-0.66, +0.82]$, $W = 40.0$, $p = 0.70$); Regime B ($\approx$ 25–30% faults) -0.04 FA/h (95% CI $[-0.78, +0.69]$, $W = 52.0$, $p = 0.98$); and F7 (Optical Noise) -0.49 FA/h (95% CI $[-1.47, +0.38]$, $W = 17.0$, $p = 0.28$). In every condition, 95% bootstrap confidence intervals spanned zero, and Wilcoxon signed-rank tests yielded $p \ge 0.28$.

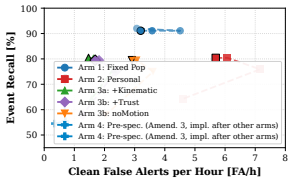
B. Secondary Benchmark Findings

Kinematic motion gating drives alert accuracy: Kinematic stillness gating accounted for the largest share of false-alert suppression: relative to ungated Arm 2, Arm 3a reduced false alerts per hour by 74.2% in clean data (1.47 vs 5.69 FA/h) and by 67.9% in Regime B (2.30 vs 7.16 FA/h) (Table IV). Furthermore, the no-motion trust ablation (Arm 3b-noMotion) alone reduced clean false alerts by about half (5.69 to 2.91 FA/h).

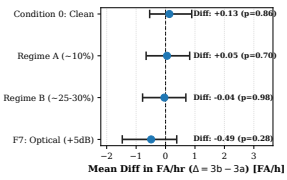
Recall degradation under optical corruption (F7): Under intense motion-coupled optical noise, Arm 3b recall degraded to 58.0% compared to 64.3% in Arm 3a; two subjects drive the F7 recall difference (exploratory). In fault-disaggregated analysis of Regime B, stale sensor values (F2) in Arm 1 and scale errors (F6) in Arm 2 produced the highest false alert rates (up to 20.2 and 36.6 FA/h, respectively); while kinematic gating suppressed stale-freeze false alerts, F2 episodes represent small exposure slices (< 1.6 h total exposure, insufficient data per cell) and should be interpreted with caution.

C. Edge-of-Grid Analysis and Fallback Rule Invocation

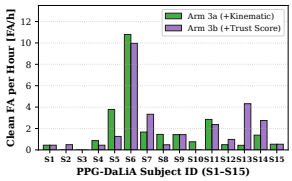
Hyperparameter tuning consistently selected parameter boundary values. Most notably, **Arm 2 never attained the $\ge 85\%$ recall target on any training fold in any condition** (maximum training recall achieved was $\approx 80.4\%$). Consequently, the fallback selection rule (maximizing training recall) was invoked in 100% (15/15) of folds for Arm 2 across all regimes, selecting $k = 2.0$ and $P = 10$ s. For Arm 1, tuning selected $\tau_{\text{pop}} = 100$ bpm and $P = 30$ s across all folds.



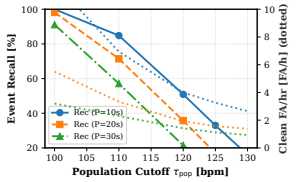
(a) Event Recall vs Clean False Alerts per Hour across Arms and Regimes



(b) Paired Differences (Δ = Arm 3b − Arm 3a) with 95% Bootstrap CIs



(c) Per-Subject False Alerts per Hour in Clean Data (Condition 0)



(d) Arm 1 Threshold Sensitivity across Cutoff τ_{pop} and Persistence P

Fig. 2. Empirical benchmark findings across arms and conditions. (a) Event recall vs clean false alerts per hour. (b) Forest plot of paired differences (Δ = Arm 3b − Arm 3a) with 95% bootstrap CIs; (c) Per-subject FA/hr in Condition 0; (d) Arm 1 sensitivity across τ_{pop} and persistence P.

TABLE IV
LEAVE-ONE-SUBJECT-OUT COMBINED ALERTING BENCHMARK RESULTS ACROSS CONDITIONS (PRECISION IS ALERT-LEVEL AND RECALL IS EPISODE-LEVEL)

Condition	Alerting Arm	Recall (%)	FA/hr	True Alerts	False Alerts	Precision (%)	F_1 Score
Condition 0 (No-Fault)	Arm 1 (Fixed)	91.1	3.20	132	100	56.9	0.700
	Arm 2 (Personal)	80.4	5.69	133	178	42.8	0.558
	Arm 3a (+Kinematic)	80.4	1.47	133	46	74.3	0.772
	Arm 3b (+Trust Score)	79.5	1.69	130	53	71.0	0.750
	Arm 3b-noMotion	79.5	2.91	130	91	58.8	0.676
	Arm 4 (pre-specified in Amendment 3, implemented after the other arms' results were seen)	72.3	0.70	113	22	83.7	0.776
Regime A (Low Duty 10%)	Arm 1 (Fixed)	91.1	3.58	132	112	54.1	0.679
	Arm 2 (Personal)	80.4	6.08	132	190	41.0	0.543
	Arm 3a (+Kinematic)	80.4	1.63	132	51	72.1	0.760
	Arm 3b (+Trust Score)	79.5	1.82	129	57	69.4	0.741
	Arm 3b-noMotion	79.5	3.07	129	96	57.3	0.666
	Arm 4 (pre-specified in Amendment 3, implemented after the other arms' results were seen)	62.5	0.67	96	21	82.1	0.710
Regime B (Nominal 25-30%)	Arm 1 (Fixed)	91.1	4.51	132	141	48.4	0.632
	Arm 2 (Personal)	75.9	7.16	125	224	35.8	0.487
	Arm 3a (+Kinematic)	75.9	2.30	125	72	63.5	0.691
	Arm 3b (+Trust Score)	75.0	2.37	122	74	62.2	0.680
	Arm 3b-noMotion	75.0	3.61	122	113	51.9	0.614
	Arm 4 (pre-specified in Amendment 3, implemented after the other arms' results were seen)	55.4	0.70	83	22	79.0	0.651
F7 (High-Motion BVP Noise)	Arm 1 (Fixed)	92.0	3.07	131	96	57.7	0.709
	Arm 2 (Personal)	64.3	4.61	100	144	41.0	0.501
	Arm 3a (+Kinematic)	64.3	1.57	100	49	67.1	0.657
	Arm 3b (+Trust Score)	58.0	1.28	90	40	69.2	0.631
	Arm 3b-noMotion	58.0	2.05	90	64	58.4	0.582
	Arm 4 (pre-specified in Amendment 3, implemented after the other arms' results were seen)	54.5	0.35	82	11	88.2	0.673

TABLE V
PAIRED STATISTICAL COMPARISON & SUBGROUP STRATIFICATION

Part A: Paired Difference (Δ = Arm 3b − Arm 3a)				
Condition	**Mean Δ**	**95% CI**	W	p-**value**
Clean (Cond 0)	$+0.13$	$[-0.54, +0.89]$	31.0	0.86
Regime A ($\approx 10\%$)	$+0.05$	$[-0.66, +0.82]$	40.0	0.70
Regime B ($\approx 25\%$)	-0.04	$[-0.78, +0.69]$	52.0	0.98
F7 Optical ($+5$ dB)	-0.49	$[-1.47, +0.38]$	17.0	0.28
Part B: Stratified Event Recall (%) / Clean FA/hr				
Stratum	**Arm 1**	**Arm 2**	**Arm 3a**	**Arm 3b**
Clean: All-15	91 / 3.2	80 / 5.7	80 / 1.5	80 / 1.7
Clean: without S5	91 / 2.9	80 / 5.3	80 / 1.4	79 / 1.7
Clean: S5/6/11	100 / 9.4	94 / 16.1	94 / 5.1	94 / 3.9
Reg B: All-15	91 / 4.5	76 / 7.2	76 / 2.3	75 / 2.4
Reg B: without S5	91 / 4.1	75 / 6.4	75 / 2.2	74 / 2.3
F7: All-15	92 / 3.1	64 / 4.6	64 / 1.6	58 / 1.3
F7: without S5	92 / 2.7	63 / 4.0	63 / 1.2	56 / 1.1

D. Arm 4 Conformal Abstention Evaluation (Pre-specified in Amendment 3, Implemented After Other Arms' Results Were Seen)

To benchmark distribution-free selective alerting, Arm 4 was pre-specified in Amendment 3, implemented after the other arms' results were seen (Draft Amendment 5). Inductive conformal calibration was performed on resting training windows with miscoverage parameter α tuned over $\{0.05, 0.10\}$ ($\alpha = 0.10$, $\lambda = 0.5$, $\theta_{\text{unc}} = 0.30$, and $P = 10$ s selected across all 15 LOSO training folds). Across all evaluated regimes, Arm 4 achieved the lowest clean false-alert frequency observed in the benchmark: 0.70 FA/h in clean data (a 52.4% reduction over Arm 3a and an 87.7% reduction over Arm 2), 0.67 FA/h in Regime A, and 0.35 FA/h under F7 optical noise (Table IV). Precision reached 83.7% in clean telemetry and 88.2% under F7 noise ($F_1 = 0.776$ clean, 0.673 F7) on the alert-count basis (Table IV); Arm 4's clean F_1 differs from Arm 3a (0.772) by 0.004 with no confidence interval and is not claimed as an improvement. However, conservative abstention came at the expense of sensitivity: episode recall dropped from 80.4% to 72.3% (81/112 episodes) in clean data, and to 54.5% (61/112 episodes) in F7 noise, demonstrating the fundamental trade-off between strict false-alert suppression and event sensitivity under non-stationary ambulatory noise.

VIII. Limitations

Methodological constraints and threats to validity are summarized in Table VI: (1) Resting tachycardia episodes were synthetically injected due to natural event sparsity, making Arm 1 recall ($> 91\%$) intrinsically dependent on the synthetic $[105, 135]$ bpm injection range relative to the 100 bpm floor; (2) Cohort is limited to 15 healthy young participants in the PPG-DaLiA protocol, leaving free-living 24/7 unconstrained monitoring unproven; (3) Front-end PPG tracking errors propagate to downstream logic; (4) Threshold cutoffs ($\tau_{\text{pop}}, \theta_{\text{rest}}, \theta_{\text{trust}}$) represent engineering design choices with unopened sources; (5) Subject S5 exhibited sustained elevated resting heart rate of unknown cause (reference ECG clean), necessitating dual pooled and stratified reporting; (6) Arm 4 was pre-specified in Amendment 3, implemented after the other arms' results were seen with inductive conformal calibration ($\alpha \in \{0.05, 0.10\}$); and (7) External datasets (PhysioNet Wearable, WESAD) were not evaluated.

TABLE VI
Methodological Limitations & Protocol Amendments (from git log)

Part A: Methodological Constraints and Threats to Validity	
Synthetic Injection	112 episodes (105–135 bpm); Arm 1 recall tied to range vs 100 bpm floor.
Cohort & Protocol	$N = 15$ healthy adults in PPG-DaLiA; free-living unconstrained monitoring unproven.
Subject S5 Outlier	Elevated resting HR; reference ECG clean; reported pooled (All-15) and without S5.
Design Choice Bounds	Cutoffs ($\tau_{\text{pop}}, \theta_{\text{rest}}, \theta_{\text{trust}}$) are design choices evaluated via sensitivity grids.
Grid Boundaries	Boundary selections: Arm 1 ($\tau_{\text{pop}} = 100$ bpm); Arm 2 ($k = 2.0$, max-recall fallback).
Part B: Protocol Amendments and Provenance (from git log)	
Protocol Freeze [41effle]	2026-10-18: Frozen protocol: H_1–H_3, faults F1–F7, LOSO CV.
ENMO Scale Fix [3a5ccc2]	2026-10-19: Removed /64 division; enforced median norm check $[0.8, 1.2]$ g.
Adaptive Ramp [e358bf1]	2026-10-19: Adapted ramp rates ($r_{\text{lo}}, r_{\text{hi}}$) clamped to $[105, 135]$ bpm.
Emitter Rules [cad4c7f]	2026-10-19: StandardAlertEmitter ($P \in \{10, 20, 30\}$ s, $R = 60$ s) across all arms.
Grid Widening [a9ce657]	2026-10-19: Widened search grids for τ_{pop} and k following boundary selection.
Fallback Selection [f92ca8d]	2026-10-19: Arm 2 recall ceiling ($\approx 80.4\%$) triggered fallback in 100% of folds.
Arm 4 Impl. [Amend. 5]	2026-10-22: Pre-specified Amend. 3, impl. after other arms ($\alpha = 0.10$ selected).

IX. Future Scope and Conclusion

Future Scope: Future efforts will extend this benchmark along five technical avenues: (1) online dynamic conformal martingales under covariate shifts [4]; (2) external wearable corpora evaluation, including PhysioNet Wearable [6] and WESAD; (3) ambulatory inpatient holter validation with spontaneous arrhythmias; (4) on-device microcontroller deployment of causal baseline and kinematic gating engines; and (5) prospective clinical validation of operational symptom review (FR-06) and local explanation (FR-07) modules.

Conclusion: In this pre-specified and version-tagged fault-injection benchmark, kinematic motion gating accounted for the largest share of false-alert suppression in wearable heart-rate monitoring, while the no-motion trust ablation (Arm 3b-noMotion) alone reduced clean false alerts by about half (5.69 to 2.91 FA/h). Multi-component software trust scoring offered no statistically detectable incremental benefit over kinematic stillness gating. Fixed thresholding with persistence achieved the highest recall but lower precision, and its recall depends directly on the 105–135 bpm injection range. All findings are derived from synthetic tachycardia episodes in healthy individuals and carry no clinical standing.

Regulatory, Data, and Code Availability

Regulatory Status (Not a Medical Device): VitaSense AI software, alerting algorithms, and benchmark suite are academic research artifacts. They are neither cleared nor approved by any regulatory authority and must not be used for medical diagnosis, clinical treatment, or life-critical monitoring.

Data and Code Availability: PPG-DaLiA is openly accessible via UCI [1], [5]; PhysioNet Wearable is under ODC-By v1.0 [6]. Benchmark code, execution scripts, and numerical provenance logs are provided in the repository.

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